Software Requirements Specification (SRS)

Integrated eBay Order, Inventory, 3PL, Warehouse and Finance Operating Platform

Version 0.1 (Draft for client review) | 29 October 2026 | Prepared from the Requirement Analysis (Part 1)

1. Document Control

1.1 Purpose

This SRS defines the functional, data, integration, non-functional and acceptance requirements for a web-based operating platform covering eBay orders, inventory, procurement, 3PL fulfilment, warehouse, shipping, finance, returns and reporting. It is the baseline for design, development, testing and client sign-off.

1.2 Conventions

- **PCA** = Pending Client Approval. The item rests on a provisional assumption from Part 1 (Q-01 to Q-24) and may change.
- Requirement IDs: `FR-<MODULE>-<nn>`. Priority: **M** Must, **S** Should, **C** Could. Phase: P1 to P5.
- Nothing in this document is an approved client decision unless it is marked as such. Budget and timeline are developer-proposal items and are not invented here.

1.3 Revision history

Version	Date	Author	Change
0.1	2026-10-29	Analyst	First full draft from Part 1 analysis

1.4 Key definitions

Term	Meaning
Owner	The party that owns a stock unit: the operator itself or a 3PL client

Term	Meaning
Client	A 3PL customer of the operator (not an eBay buyer)
Customer	An end buyer (for example an eBay buyer)
Stock ledger	Append-only record of every stock movement
Financial ledger	Append-only record of every financial posting
Marketplace adapter	Interface isolating eBay-specific logic from the core
On hand / Reserved / Available	Physical quantity / quantity committed to orders / on hand minus reserved (see S7)

2. Business Context

2.1 Overview

The client operates an e-commerce and fulfilment business and needs one integrated platform instead of separate tools. The platform must connect eBay orders, stock, supplier purchasing, 3PL client services, warehouse locations, shipping, invoicing, payments, expenses, returns and profitability so that operational and financial reports reconcile (client Section 30).

2.2 Objectives and proposed KPIs (PCA)

The client gave nine objectives without measures. Proposed measures for approval:

#	Objective	Proposed KPI
O1	Single view of orders across eBay accounts	100% of eBay orders imported within 15 minutes of creation
O2	Accurate stock	Ledger-to-physical variance under 1% at cycle count
O3	Control of procurement	100% of stock receipts linked to a PO or a reasoned manual receipt

#	Objective	Proposed KPI
O4	3PL client service	Client portal shows only that client's data, zero cross-client leaks
O5	Warehouse traceability	Every unit traceable to a location and a movement
O6	Shipping visibility	Every dispatched order has a carrier and tracking reference
O7	Financial control	Invoice, payment and expense totals reconcile to the ledger, zero unexplained difference
O8	Profit visibility	Gross and net profit per order, SKU and client
O9	Reporting	The 14 required reports available with export

2.3 Assumptions and constraints

ID	Statement	Status
AS-01	The operator sells own stock and fulfils for 3PL clients; each stock record has an owner	PCA (Q-01)
AS-02	One base currency, configurable timezone	PCA (Q-08)
AS-03	Payments are recorded manually; no payment gateway	PCA (Q-18)
AS-04	eBay integration is read-only in the MVP	PCA (Q-07)
AS-05	Existing data migration is out of scope until confirmed	PCA (Q-15)
CON-01	eBay access depends on an approved developer account and credentials, to be checked against current official eBay documentation	Dependency D-01

3. Scope and Phases

3.1 Phase plan

Part 1 found modules without a phase (A-01 to A-04). The plan below resolves them provisionally. All placements are PCA.

Phase	Content
P1	Auth and roles, audit log, eBay order import (adapter), products and SKU mapping, inventory with ledger (single default location), stock adjustments, order management, basic dashboard, basic reports, in-app alert indicators
P2	Suppliers, purchase orders and receiving, returns and refund records, customers, warehouse locations, bundles/kits if confirmed
P3	3PL clients, client portal, fulfilment workflow, storage rate cards and billing
P4	Invoices, payments, expenses, profitability, financial reports
P5	Notification engine, advanced reports, carrier API integration, eBay write-back if approved

Dashboard KPIs that depend on later modules (Supplier Payables, Client Receivables, Storage Revenue, Outstanding Payments) are hidden until the module exists (resolves A-02). Low-stock indicators appear on the dashboard in P1; the notification engine is P5 (resolves A-03).

3.2 Out of scope (client Section 29, future)

Additional marketplaces, label printing, barcode and handheld scanning, mobile app, multi-company, payment gateway, automated accounting export, and AI forecasting. The architecture must not block these, but none is built in the MVP.

4. User Roles and Permission Matrix

4.1 Roles

Super Admin, Operations Manager, Warehouse Staff, Finance User, 3PL Client User, Read-only/Viewer (the client's six roles). Permissions below are proposed in full because the client gave only one-line summaries (PCA).

4.2 Matrix

C create, R read, U update, D delete or void, A approve, X export, - none. Client users see only their own client records.

Module	Super Admin	Ops Manager	Warehouse	Finance	3PL Client	Viewer
Users and roles	CRUDA	R	-	-	-	-
System configuration	CRUDA	R	-	R	-	-
Dashboard	R	R	R (ops)	R (finance)	R (own)	R
eBay accounts	CRUD	R	-	-	-	-
Orders	CRUDX	CRUAX	RU (status)	R	R (own)	R
Products and SKUs	CRUD	CRUD	R	R	R (own)	R
Inventory and ledger	R X	RAX	RU	R	R (own)	R
Stock adjustments	CRA	CRA	C (request)	R	-	-
Suppliers and POs	CRUD	CRUA	R	R	-	-
Receiving	R	CRU	CRU	R	-	-
3PL clients and rates	CRUD	CRUD	R	R	R (own)	-
Fulfilment	R	CRUA	CRU	R	R (own)	R
Warehouse and locations	CRUD	CRUD	RU	-	-	R
Shipping	R	CRU	CRU	R	R (own)	R
Invoices	CRUDX	R	-	CRUAX	R (own)	-
Payments	CRUDX	R	-	CRUX	R (own)	-
Expenses	CRUD	R	-	CRUAX	-	-
Profitability	R X	R X	-	R X	R (own)	-
Returns	CRUD	CRUA	CRU	R	R (own)	R
Reports	RX	RX	R (ops)	RX (finance)	R (own)	R
Audit log	R X	R	-	R (finance)	-	-

Rules: voiding financial records requires Super Admin and a reason; Finance cannot alter stock; Warehouse cannot see costs or margins.

5. Functional Requirements

5.1 FR-AUTH Authentication and Access (P1)

ID	Requirement	Pri
FR-AUTH-01	Users sign in with email and password; passwords stored with a salted adaptive hash	M
FR-AUTH-02	Role-based access control enforced server-side on every endpoint and query	M
FR-AUTH-03	Session expiry after a configurable idle time (default 30 minutes) and logout	M
FR-AUTH-04	Account lockout after configurable failed attempts, admin unlock	M
FR-AUTH-05	Password reset by time-limited single-use link	M
FR-AUTH-06	Two-factor authentication for admin and finance roles	S
FR-AUTH-07	3PL client users are bound to one client; every query is filtered by client id on the server	M

5.2 FR-DASH Dashboard (P1, extended later)

ID	Requirement	Pri
FR-DASH-01	Show the 10 client KPIs using definitions in S8.3; widgets for modules not yet released are hidden	M
FR-DASH-02	Charts: orders per day, revenue, stock by status, top SKUs	S
FR-DASH-03	Date range and eBay account filters	M
FR-DASH-04	Quick actions: new adjustment, view pending orders, view low stock	S
FR-DASH-05	Each KPI links to a filtered list that reproduces the number	M

ID	Requirement	Pri
FR-DASH-06	3PL clients see a dashboard limited to their own data	M (P3)

5.3 FR-EBAY eBay Integration (P1)

ID	Requirement	Pri
FR-EBAY-01	Connect one or more eBay seller accounts through OAuth; schema supports many from day one (A-05, PCA)	M
FR-EBAY-02	All eBay logic sits behind a marketplace adapter interface (connect, fetch orders, fetch listings, map status)	M
FR-EBAY-03	Import orders by scheduled polling, with an optional notification trigger if available	M
FR-EBAY-04	Imports are idempotent using the eBay order id and account id as a unique key	M
FR-EBAY-05	Store the raw eBay payload with each order for traceability	M
FR-EBAY-06	Map eBay status to the canonical order status (S7.1)	M
FR-EBAY-07	Map eBay listing SKU to internal SKU; unmapped lines go to an exception queue	M
FR-EBAY-08	Handle rate limits with back-off and resume; token refresh handled automatically	M
FR-EBAY-09	Show per-account sync status, last success, and errors	M
FR-EBAY-10	Write-back (tracking, shipped, cancel) is not in the MVP (PCA, Q-07)	C (P5)

Compatibility with eBay APIs must be verified against current official documentation before design freeze (D-01).

5.4 FR-ORD Order Management (P1)

ID	Requirement	Pri
FR-ORD-01	List, search and filter orders by status, account, date, SKU, buyer and tracking	M
FR-ORD-02	Order detail with lines, buyer, address, payment state, status history and linked stock movements	M
FR-ORD-03	Canonical statuses: New, Confirmed, Allocated, Picked, Packed, Shipped, Delivered, Cancelled, Returned (PCA, A-09)	M
FR-ORD-04	Stock is reserved when the order reaches Confirmed and deducted at dispatch (PCA, Q-02)	M
FR-ORD-05	Insufficient stock marks the line Backordered and raises an indicator; the order is not silently dropped	M
FR-ORD-06	Status changes are manual or workflow-driven, validated against allowed transitions, and logged with user and time	M
FR-ORD-07	Cancellation releases reservations; cancellation after dispatch routes to Returns	M
FR-ORD-08	Manual order creation for non-eBay sales, flagged with its source	S
FR-ORD-09	Export of the filtered list to CSV and Excel	M

5.5 FR-PRD Products and SKUs (P1)

ID	Requirement	Pri
FR-PRD-01	Maintain products with internal SKU, name, barcode, dimensions, weight, owner, status and cost	M
FR-PRD-02	One internal SKU maps to many marketplace SKUs or listings (D-03)	M

ID	Requirement	Pri
FR-PRD-03	SKU is unique per owner; SKUs cannot be hard-deleted once used, only deactivated	M
FR-PRD-04	Bundles/kits with component quantities, deferred until confirmed (Q-23)	C (P2)
FR-PRD-05	Bulk import and export of products by CSV	S

5.6 FR-INV Inventory (P1)

ID	Requirement	Pri
FR-INV-01	Every stock change is a stock ledger entry (append-only) with SKU, owner, location, quantity delta, state, source document and user	M
FR-INV-02	Stock states: On hand, Reserved, Available, Incoming, Damaged, Quarantined, Missing (definitions in S7.2, PCA)	M
FR-INV-03	Current balances are derived from, and reconcilable to, the ledger	M
FR-INV-04	P1 uses one default location; the schema supports bin-level locations from the start (A-04)	M
FR-INV-05	Stock view by SKU, location, owner and state, with ledger drill-down	M
FR-INV-06	Valuation by weighted average cost (PCA, Q-06)	M (P2 for cost)
FR-INV-07	Low-stock threshold per SKU with a dashboard indicator	M
FR-INV-08	Cycle count with variance posting through an approved adjustment	S
FR-INV-09	Negative on-hand stock is not allowed unless a configuration permits it	M

5.7 FR-ADJ Stock Adjustments (P1)

ID	Requirement	Pri
FR-ADJ-01	Adjustment requires SKU, location, quantity, reason code and free-text note	M
FR-ADJ-02	Adjustments above a configurable quantity or value need Manager approval (PCA, Q-13)	M
FR-ADJ-03	Approved adjustments post to the ledger; rejected ones are retained with the reason	M
FR-ADJ-04	The requester cannot approve their own adjustment	M
FR-ADJ-05	Reason codes: damage, loss, found, count correction, quarantine, write-off	M

5.8 FR-SUP Suppliers (P2)

ID	Requirement	Pri
FR-SUP-01	Supplier records with contact, terms, currency and lead time	M
FR-SUP-02	Supplier-product links with supplier SKU and last price	S
FR-SUP-03	Supplier invoices and amounts payable per supplier	M
FR-SUP-04	Supplier payments recorded against supplier invoices (resolves B-14, PCA)	M (P4)

5.9 FR-PO Purchase Orders and Receiving (P2)

ID	Requirement	Pri
FR-PO-01	PO lifecycle: Draft, Approved, Sent, Partially received, Received, Closed, Cancelled	M

ID	Requirement	Pri
FR-PO-02	Receiving creates ledger entries; partial and over-receipts allowed within a tolerance	M
FR-PO-03	Receiving discrepancies (short, damaged) recorded with reason	M
FR-PO-04	Landed costs allocated to received lines for valuation	S
FR-PO-05	Incoming stock is shown from approved POs	S

5.10 FR-3PL Clients (P3)

ID	Requirement	Pri
FR-3PL-01	Client records with contact, billing details, rate card and status	M
FR-3PL-02	Every product, stock line, order and charge carries a client owner	M
FR-3PL-03	Client portal with read access to own stock, orders and invoices	M
FR-3PL-04	Client inbound notices (expected deliveries)	S
FR-3PL-05	Client pricing rules are configurable: pick, pack, handling, storage, special services (PCA)	M

5.11 FR-FUL Fulfilment (P3)

ID	Requirement	Pri
FR-FUL-01	Flow: Received, Allocated, Picked, Packed, Shipped, mapped to canonical statuses (A-09)	M
FR-FUL-02	Pick lists by location and order	M
FR-FUL-03	Pack confirmation records items, parcels and weight	M

ID	Requirement	Pri
FR-FUL-04	Service charges are generated from rate cards when work is completed	M

5.12 FR-WH Warehouse and Locations (P2)

ID	Requirement	Pri
FR-WH-01	Warehouses, zones, aisles and bins in a hierarchy	M
FR-WH-02	Stock transfer between locations posts paired ledger entries	M
FR-WH-03	Location capacity and type (pallet, shelf, bin) attributes	S
FR-WH-04	One warehouse in the UI at launch; schema supports many (PCA, Q-05)	M

5.13 FR-STO Storage Billing (P3)

ID	Requirement	Pri
FR-STO-01	Configurable rate cards by basis: pallet, shelf, bin, volume or item (PCA, Q-11)	M
FR-STO-02	Periodic snapshots of occupied storage per client	M
FR-STO-03	Billing run with proration and minimum charge options, previewed before posting	M
FR-STO-04	Storage revenue posts to the financial ledger and invoice lines	M

5.14 FR-SHIP Shipping (P1 manual, P5 carriers)

ID	Requirement	Pri
FR-SHIP-01	Record carrier, service, tracking number, weight and cost manually	M
FR-SHIP-02	Tracking link generated per carrier	S
FR-SHIP-03	Dispatch posts the stock deduction (links to FR-ORD-04)	M
FR-SHIP-04	Carrier API integration behind a carrier adapter interface (D-07)	C (P5)

5.15 FR-CUS Customers (P2)

ID	Requirement	Pri
FR-CUS-01	Customers (buyers) are stored separately from 3PL clients (PCA, Q-16)	M
FR-CUS-02	Customer records link to orders, returns and invoices	M
FR-CUS-03	Personal data handled under the privacy rules in S14	M

5.16 FR-INVC Invoices (P4)

ID	Requirement	Pri
FR-INVC-01	Invoice types: 3PL service, storage, supplier, other sales (eBay orders are sales records, not invoices, PCA Q-09)	M
FR-INVC-02	Sequential, gap-free numbering per series	M
FR-INVC-03	VAT calculation by configurable rates and rules (PCA, Q-08)	M

ID	Requirement	Pri
FR-INVC-04	Issued invoices are immutable; corrections use a reversing entry	M
FR-INVC-05	PDF output and email-ready download	S

5.17 FR-PAY Payments (P4)

ID	Requirement	Pri
FR-PAY-01	Manual payment recording: Bank transfer, Card, Cash, Other (PCA, Q-18)	M
FR-PAY-02	Allocation of one payment to many invoices and the reverse	M
FR-PAY-03	Overpayment held as credit on account	S
FR-PAY-04	Invoice status derived: Draft, Issued, Part paid, Paid, Overdue, Void	M

5.18 FR-EXP Expenses (P4)

ID	Requirement	Pri
FR-EXP-01	Expense entry with category, supplier, date, amount, VAT and attachment	M
FR-EXP-02	Configurable categories and a flag for allocation to orders, SKUs or clients	M
FR-EXP-03	Approval for expenses above a threshold	S

5.19 FR-PRF Profitability (P4)

ID	Requirement	Pri
FR-PRF-01	Gross profit = revenue less COGS, marketplace fees and shipping cost (PCA, Q-12)	M

ID	Requirement	Pri
FR-PRF-02	Net operating profit = gross profit less allocated expenses	M
FR-PRF-03	Profit by order, SKU, client and period, with drill-down to source postings	M
FR-PRF-04	Cost fields are configurable and each report states which are included	M

5.20 FR-RET Returns and Refunds (P2)

ID	Requirement	Pri
FR-RET-01	Return created against an order with reason code	M
FR-RET-02	Inspection results in parallel dispositions: Restock, Damaged, Quarantine, Dispose (PCA, B-08)	M
FR-RET-03	Disposition posts the matching ledger entries	M
FR-RET-04	Refund record with amount and reason; formal credit notes are later (PCA, A-07, Q-24)	M

5.21 FR-RPT Reports (P1 basic, P5 advanced)

ID	Requirement	Pri
FR-RPT-01	The 14 client reports, each with defined columns, filters and a calculation note (S15)	M
FR-RPT-02	Export to CSV and Excel; PDF for selected reports	M
FR-RPT-03	Report totals reconcile to the ledgers and any difference is shown	M

5.22 FR-NOT Notifications (P5, in-app indicators in P1)

ID	Requirement	Pri
FR-NOT-01	The nine client alert types with configurable thresholds (S16)	M
FR-NOT-02	In-app notification centre first; email later (PCA, Q-22)	M
FR-NOT-03	Users can acknowledge and mute by type	S

5.23 FR-SRCH Search, Filters and Export (P1)

ID	Requirement	Pri
FR-SRCH-01	Global search across orders, SKUs, customers and tracking numbers	M
FR-SRCH-02	Saved filters per user	C
FR-SRCH-03	Export of any list honors the user's permissions and filters	M

5.24 FR-AUD Audit Log (P1)

ID	Requirement	Pri
FR-AUD-01	Log who, when, what, before and after values for create, update, delete, approve and export actions	M
FR-AUD-02	Logs are append-only and cannot be edited through the application	M
FR-AUD-03	Retention period configurable; default proposed in S14 (PCA, Q-20)	M

5.25 FR-CFG Configuration (P1 onward)

ID	Requirement	Pri
FR-CFG-01	Commercial rules (rates, thresholds, VAT, cost fields, numbering) are configuration, not code	M
FR-CFG-02	Configuration changes are audited and effective-dated where they affect pricing	M

5.26 FR-INT API and Integrations (P1 onward)

ID	Requirement	Pri
FR-INT-01	Internal REST API used by the UI, with the same authorization as the UI	M
FR-INT-02	Adapter interfaces for marketplace and carrier integrations	M
FR-INT-03	Outbound webhooks for order and stock events	C

5.27 FR-OPS Operations and Support (P1 onward)

ID	Requirement	Pri
FR-OPS-01	Seed and demo data are clearly labeled and never presented as evidence of a working integration (C-08)	M
FR-OPS-02	Health checks, background job monitoring and an exception queue view	M
FR-OPS-03	Automated daily backups with a tested restore procedure	M

6. Workflows

Each workflow is stated as ordered steps with the state change and ledger effect.

6.1 eBay order to dispatch

1. Scheduler calls the adapter and imports new orders (idempotent). State: **New**.
2. Lines are mapped to internal SKUs. Unmapped lines go to the exception queue.
3. When the order is paid or confirmed, the system reserves stock. State: **Confirmed**, then **Allocated**. Ledger: reservation entry.
4. Staff pick and pack. States: **Picked**, **Packed**.
5. Tracking is recorded and the order is dispatched. State: **Shipped**. Ledger: stock deduction.
6. Carrier confirmation or timeout moves the order to **Delivered**.

6.2 Purchase order to stock

1. PO drafted, approved, sent. Stock shows as Incoming.
2. Goods are received against the PO. Ledger: receipt entries, discrepancies recorded.
3. Supplier invoice is matched to the receipt. Payable is created in the financial ledger.
4. Payment recorded against the invoice and the payable is cleared.

6.3 Return to disposition

1. Return opened with reason. Order state **Returned** only after receipt.
2. Goods inspected. Each unit receives a disposition: Restock, Damaged, Quarantine or Dispose.
3. Ledger entries post per disposition. Refund record is created if applicable.

6.4 Stock adjustment approval

1. Requester submits the adjustment. If under the threshold, it posts automatically.
2. Otherwise a Manager approves or rejects. The requester cannot approve their own.
3. Approved adjustments post to the ledger. All outcomes are audited.

6.5 3PL storage billing

1. A snapshot job records occupied storage per client.
2. A billing run applies the rate card, proration and minimum charge.
3. Finance reviews the preview, then posts. Invoice lines and the financial ledger are created together.

7. Business Rules

7.1 Canonical order status transitions (PCA)

From	Allowed to
New	Confirmed, Cancelled
Confirmed	Allocated, Cancelled
Allocated	Picked, Cancelled
Picked	Packed, Allocated (re-pick), Cancelled
Packed	Shipped, Cancelled
Shipped	Delivered, Returned
Delivered	Returned
Cancelled, Returned	none (terminal)

eBay status mapping is held in a configurable table and any unknown eBay status goes to the exception queue.

7.2 Stock state definitions (PCA, B-02)

State	Definition
On hand	Physical units in the warehouse, in sellable or non-sellable condition
Reserved	On-hand units committed to confirmed orders
Available	On-hand sellable units minus Reserved
Incoming	Units on approved POs or client inbound notices, not yet received
Damaged	On hand but not sellable, awaiting disposal or repair
Quarantined	On hand but held pending inspection

State	Definition
Missing	Units the ledger shows but a count did not find, pending investigation
Total stock	On hand including Damaged and Quarantined, excluding Missing and Incoming

7.3 Decision table for allocation

Available covers line	Backorder allowed	Result
Yes	n/a	Reserve full quantity
Partly	Yes	Reserve available, mark remainder Backordered
Partly	No	Reserve nothing, flag the order for manual action
No	Yes	Mark Backordered
No	No	Flag for manual action

7.4 Other rules

- **BR-01** Ledgers are append-only. Corrections are new reversing entries.
- **BR-02** Every financial and stock posting links to a source document.
- **BR-03** An owner's stock cannot be allocated to another owner's order.
- **BR-04** Voided or cancelled documents keep their numbers.
- **BR-05** Issued invoices cannot be edited, only reversed.
- **BR-06** Date and time are stored in UTC and shown in the configured business timezone.

8. UI/UX and KPI Definitions

8.1 Design principles

- Web-based, responsive for desktop, laptop and tablet. Tablet supports full warehouse operations; phone is limited to read-only views and simple status updates (PCA, A-08, Q-19).
- Consistent list pattern: search, filters, sortable columns, pagination, export.
- Every number on a dashboard or report links to the records behind it.
- Destructive and financial actions need confirmation and a reason.
- WCAG 2.1 AA as the target for accessibility, English as the only language in the MVP (PCA, C-07).

8.2 Main navigation

The client's 20 navigation items are kept. Items belonging to later phases are hidden until released, and items are hidden per role by the matrix in S4.

Group	Items
Overview	Dashboard
Sales	Orders, eBay Accounts, Customers, Returns
Stock	Products, Inventory, Adjustments, Warehouse, Locations
Buying	Suppliers, Purchase Orders
Fulfilment	3PL Clients, Fulfilment, Shipping, Storage
Finance	Invoices, Payments, Expenses, Profitability
System	Reports, Notifications, Audit Log, Settings

8.3 KPI definitions (PCA, Q-10)

KPI	Proposed definition	Available from
Total Orders	Count of orders created in the selected period, excluding Cancelled	P1
Pending Orders	Orders in New, Confirmed, Allocated, Picked or Packed	P1
Revenue	Sum of order item totals excluding tax in the period	P1
Low Stock Items	SKUs with Available at or below their threshold	P1
Total Stock Units	Total stock as defined in S7.2	P1
Inventory Value	On-hand quantity times weighted average cost	P2
Supplier Payables	Unpaid supplier invoice balances	P4
Client Receivables	Unpaid client invoice balances	P4
Storage Revenue	Storage charges invoiced in the period	P4
Outstanding Payments	Receivables past due date	P4

9. API and Integration Specification

9.1 Internal API

- REST over HTTPS with JSON, versioned under /api/v1. Authentication by session token or bearer token.
- Authorization checks on every call, including owner and client filtering.
- Standard error envelope: code, message, field errors and a correlation id.
- List endpoints support pagination, filtering and sorting. Mutating endpoints accept an idempotency key.

9.2 Resource groups

Group	Examples
Auth	login, logout, refresh, password reset
Orders	list, detail, status change, cancel
Products and SKUs	CRUD, marketplace mappings
Inventory	balances, ledger entries, adjustments, approvals
Purchasing	suppliers, POs, receipts
Fulfilment and shipping	pick lists, pack, dispatch
Finance	invoices, payments, expenses, ledger postings
Reports	run, export
Admin	users, roles, configuration, audit log

9.3 Marketplace adapter

Interface: `connect(account)`, `refreshToken(account)`, `fetchOrders(since)`, `fetchOrder(id)`, `fetchListings(page)`, `mapStatus(raw)`. The eBay adapter is the first implementation. Core services never call eBay directly.

9.4 eBay integration rules

- OAuth with stored, encrypted refresh tokens per account.
- Polling with a high-water mark per account; overlapping windows are safe because imports are idempotent.
- Back-off on rate limits and temporary errors; failures are visible on the account page.
- The raw payload is stored. Field mappings are documented once the approved API version is confirmed.
- Specific endpoints, scopes and limits must be checked against current eBay documentation before build (D-01).

9.5 Carrier adapter (P5)

Interface for rate, label, tracking and cancel. Not built in the MVP; the manual shipping fields in FR-SHIP-01 map onto it.

10. Data Model

10.1 Principles

- Every owned record has `owner_type` and `owner_id` (operator or client) so isolation does not depend on later schema changes.
- Ledgers are append-only tables with no update or delete permission for the application role.
- Surrogate keys, UTC timestamps, soft deactivation instead of deletion for master data.
- Money stored as integer minor units with a currency code.

10.2 Entity overview

Domain	Main entities
Identity	User, Role, Permission, Session
Marketplace	MarketplaceAccount, SyncRun, ExceptionItem
Sales	Order, OrderLine, OrderStatusHistory, Customer, Shipment, Parcel
Catalog	Product, Sku, MarketplaceSkuMap, Bundle
Stock	Warehouse, Location, StockLedger, StockBalance, Reservation, Adjustment, CycleCount
Purchasing	Supplier, PurchaseOrder, PoLine, Receipt, ReceiptLine
3PL	Client, RateCard, RateRule, StorageSnapshot, ServiceCharge
Finance	Invoice, InvoiceLine, Payment, PaymentAllocation, Expense, LedgerAccount, FinancialLedger
Returns	Return, ReturnLine, Disposition, Refund

Domain	Main entities
System	AuditLog, Notification, ConfigItem, Attachment

10.3 Key relationships

- Order 1..n OrderLine; OrderLine n..1 Sku; OrderLine 0..n Reservation.
- StockLedger n..1 Sku, Location, Owner; every row references a source document type and id.
- Invoice 1..n InvoiceLine; Payment n..n Invoice through PaymentAllocation.
- Return n..1 Order; ReturnLine 1..n Disposition.
- FinancialLedger rows reference the source document that caused them (invoice, payment, expense, receipt, shipment).

10.4 Core data dictionary (extract)

Table	Column	Type	Notes
stock_ledger	id	bigint	Primary key, append-only
stock_ledger	sku_id	fk	Required
stock_ledger	location_id	fk	Required
stock_ledger	owner_id	fk	Operator or client
stock_ledger	qty_delta	integer	Signed
stock_ledger	state	enum	Per S7.2
stock_ledger	source_type, source_id	text, bigint	Link to document
stock_ledger	created_by, created_at	fk, timestamptz	UTC
order	marketplace_account_id, external_order_id	fk, text	Unique together
order	status	enum	Canonical status
order	raw_payload	jsonb	Original eBay data

Table	Column	Type	Notes
financial_ledger	account_id, debit_minor, credit_minor	fk, bigint	Balanced per posting
financial_ledger	source_type, source_id	text, bigint	Link to document
audit_log	actor_id, action, entity, before, after, at	mixed	Append-only

The full dictionary is produced in the design phase from this baseline.

11. Transaction Safety

These are recommended technical solutions, labeled provisional because the client set no rules (S11 is a gap in the client document).

Risk	Control
Double import of an eBay order	Unique key on account and external order id; idempotent upsert
Overselling under concurrency	Reserve in a database transaction with row-level locking or an atomic conditional update on the balance; fail and flag if insufficient
Duplicate form submission	Idempotency key on mutating requests
Partial postings	One database transaction covers the document, ledger rows and balance change
Lost updates on edits	Optimistic locking with a version column
Ledger tampering	No update or delete grants; corrections by reversing entries
Drift between balance and ledger	Nightly job recomputes balances from the ledger and raises an alert on any difference
Failed background jobs	Retries with back-off, dead-letter queue and a visible exception list
Deadlocks	Consistent lock ordering by SKU id and location id

12. Finance Rules (provisional, PCA)

12.1 Accounting policy

- Single base currency with exchange rates stored if foreign amounts are recorded later (Q-08).
- Inventory valued at weighted average cost, recalculated on each receipt (Q-06).
- COGS is recognized at dispatch using the average cost at that time.
- Revenue from eBay orders is recognized at dispatch; marketplace fees are recorded as costs of sale.

12.2 Posting rules

Event	Debit	Credit
Goods received	Inventory	Goods received not invoiced
Supplier invoice matched	Goods received not invoiced	Supplier payable
Supplier payment	Supplier payable	Bank or cash
Order dispatched	COGS	Inventory
Order revenue recognized	Marketplace receivable	Sales revenue
Client invoice issued	Client receivable	Service and storage revenue, VAT payable
Client payment	Bank or cash	Client receivable
Expense recorded	Expense account	Payable or bank
Refund	Sales returns	Marketplace receivable or bank

12.3 Reconciliation

- Inventory value on the dashboard equals the sum of the stock ledger valued at average cost, and equals the inventory account balance.
- Receivables and payables reports equal their control account balances.
- Any mismatch is shown on a reconciliation report and raised as an alert.

13. Non-Functional Requirements

Targets are proposed because the client gave none (C-01) and are PCA pending volume figures (Q-14).

ID	Area	Target
NFR-01	Page response	95% of list and detail pages under 2 seconds at 20 concurrent users
NFR-02	Report run	Standard reports under 10 seconds for 12 months of data
NFR-03	Order import	New orders visible within 15 minutes; 5,000 orders per day supported
NFR-04	Availability	99.5% monthly excluding planned maintenance
NFR-05	Backup	Daily backups, recovery point 24 hours, recovery time 8 hours
NFR-06	Scalability	100,000 SKUs, 10 million ledger rows without redesign
NFR-07	Browsers	Current and previous major version of Chrome, Edge, Safari, Firefox
NFR-08	Maintainability	Automated tests for all ledger and posting logic; documented deployment
NFR-09	Observability	Structured logs, error tracking, job and sync dashboards
NFR-10	Data integrity	Zero unreconciled ledger differences in the nightly check

14. Security and Privacy

ID	Requirement
SEC-01	HTTPS everywhere; TLS 1.2 or higher

ID	Requirement
SEC-02	Passwords hashed with a salted adaptive algorithm; no plain storage
SEC-03	Server-side authorization and client isolation on every query, covered by automated tests
SEC-04	eBay tokens and secrets encrypted at rest; secrets kept outside source control
SEC-05	Protection against injection, cross-site scripting and request forgery; input validation server-side
SEC-06	Audit log of sensitive actions and of all exports
SEC-07	Personal data minimized to what orders and shipping need; access limited by role
SEC-08	Audit log retention of 7 years and financial records of 7 years, subject to client and legal confirmation (PCA, Q-20)
SEC-09	Data protection requirements (for example GDPR) confirmed with the client; support for access and erasure requests where law applies
SEC-10	Hosting region and provider to be agreed (Q-21)

15. Reports

Columns and calculations are proposed because the client did not define them (PCA). All reports support date filters, export and drill-down.

#	Report	Key columns and calculation	Phase
R-01	Sales by period	Orders, units, revenue by day, week, month, account	P1
R-02	Orders by status	Count and age per canonical status	P1
R-03	Stock on hand	SKU, location, owner, state, quantity	P1

#	Report	Key columns and calculation	Phase
R-04	Stock movement	Ledger entries with source document	P1
R-05	Low stock	Available against threshold	P1
R-06	Adjustments	Reason, quantity, value, approver	P1
R-07	Inventory valuation	Quantity times average cost, by owner	P2
R-08	Purchase orders and receiving	Ordered, received, outstanding, discrepancies	P2
R-09	Returns	Reason, disposition, refund value	P2
R-10	3PL client activity	Orders, pick and pack counts, charges per client	P3
R-11	Storage billing	Occupancy, rate, charge per client and period	P3
R-12	Receivables and payables ageing	Open balances by age bucket	P4
R-13	Profitability	Revenue, COGS, fees, shipping, gross and net profit	P4
R-14	Reconciliation	Ledger against balances and control accounts, differences	P4

16. Errors and Notifications

16.1 Error handling

Case	Behavior
Validation error	Field-level messages, nothing saved

Case	Behavior
Permission denied	Clear message and an audit entry for repeated attempts
Concurrency conflict	User told the record changed and offered a reload
eBay failure	Retry with back-off, show on account page, raise alert after repeated failure
Unmapped SKU or unknown status	Exception queue with a resolve action
Insufficient stock at allocation	Backorder or manual action per S7.3
Unexpected error	Generic message with a correlation id; details logged

16.2 Alert types (PCA)

The client named nine alert types without listing them. Proposed set:

#	Alert	Default trigger
N-01	Low stock	Available at or below threshold
N-02	Out of stock	Available equals zero
N-03	eBay sync failure	Three consecutive failed runs
N-04	Unmapped SKU	Any order line unmapped
N-05	Pending order too long	Confirmed for more than 24 hours
N-06	Adjustment awaiting approval	Pending for more than 24 hours
N-07	Overdue invoice	Past due date
N-08	Ledger reconciliation difference	Any difference
N-09	Return awaiting inspection	Received for more than 48 hours

17. Testing

17.1 Strategy

- Unit tests for stock, valuation, posting and status transition logic.
- Integration tests against a mocked eBay adapter and, once credentials exist, the eBay sandbox.
- Security tests for role and client isolation on every endpoint.
- Concurrency tests for simultaneous reservations and duplicate imports.
- User acceptance testing by the client per phase.

17.2 Acceptance test cases (from the client's 14 criteria, to be aligned one to one at sign-off)

ID	Case	Expected result
AT-01	Import the same eBay order twice	One order exists
AT-02	Confirm an order with sufficient stock	Reservation posted, Available reduced
AT-03	Confirm an order with insufficient stock	Backorder flag, no negative stock
AT-04	Dispatch an order	Stock deducted once, COGS posted
AT-05	Submit an adjustment above threshold	Pending until a different Manager approves
AT-06	Receive a partial PO	Stock and Incoming updated correctly
AT-07	Client user requests another client's order	Denied, nothing leaked
AT-08	Run storage billing preview and post	Invoice lines and ledger entries match
AT-09	Record a payment across two invoices	Statuses and balances correct
AT-10	Process a return with mixed dispositions	Ledger entries match each disposition

ID	Case	Expected result
AT-11	Run each of the 14 reports and export	Totals match source lists
AT-12	Run reconciliation	Zero differences on test data
AT-13	Attempt to edit an issued invoice or ledger row	Blocked; reversal available
AT-14	View audit log for a changed record	Before and after values present

Demo data results are reported as demo results and never as evidence of a working eBay integration (C-08).

18. Traceability Matrix

Client section	SRS requirements	Acceptance test
3 User roles	S4, FR-AUTH	AT-07
4 Dashboard	FR-DASH, S8.3	AT-11
5 eBay and orders	FR-EBAY, FR-ORD	AT-01, AT-02, AT-03, AT-04
6 Inventory	FR-INV, FR-ADJ, S7.2	AT-05
7 Suppliers	FR-SUP, FR-PO	AT-06
8 3PL	FR-3PL, FR-FUL	AT-07
9 Warehouse and storage	FR-WH, FR-STO	AT-08
10 Shipping	FR-SHIP	AT-04
11 Invoices and payments	FR-INVC, FR-PAY, S12	AT-09, AT-13
12 Expenses and profit	FR-EXP, FR-PRF	AT-12
13 Customers	FR-CUS	AT-07
14 Returns	FR-RET	AT-10
15 Reports	FR-RPT, S15	AT-11
16 Notifications	FR-NOT, S16	AT-12
17 Search and export	FR-SRCH	AT-11

Client section	SRS requirements	Acceptance test
18 Audit and security	FR-AUD, S14	AT-13, AT-14
19 to 24 Architecture, API, UI	S8, S9, S10, S13	NFR checks
25 Phases	S3, S19	Phase sign-off
28 Developer notes	FR-CFG	Configuration review
30 Reconciliation	S10, S11, S12	AT-12

19. Development Plan

Durations, team size and cost are developer-proposal items and are not set in this SRS. The sequence and gates below are proposed.

Phase	Main deliverables	Exit gate
P0 Preparation	Clarifications Q-01 to Q-06 answered, eBay credentials, design freeze	Signed SRS v1.0
P1 Core operations	Auth, audit, eBay import, products, inventory ledger, orders, adjustments, basic dashboard and reports	AT-01 to AT-05, AT-07, AT-14
P2 Procurement and returns	Suppliers, POs, receiving, locations, customers, returns	AT-06, AT-10
P3 3PL	Clients, portal, fulfilment, storage billing	AT-07, AT-08
P4 Finance	Invoices, payments, expenses, profitability, reconciliation	AT-09, AT-12, AT-13
P5 Alerts and advanced	Notification engine, advanced reports, carrier APIs, eBay write-back if approved	Client acceptance

20. Risks and Open Questions

20.1 Risks

ID	Risk	Impact	Mitigation
RK-01	eBay access or API changes delay import	High	Early credential request, adapter isolation, sandbox testing
RK-02	Ownership model answered late (Q-01)	High	Owner field in schema from day one
RK-03	Valuation or VAT rules change late	Medium	Configurable rules and effective dating
RK-04	Scope creep from future items (S3.2)	Medium	Change control; future list kept out of the MVP
RK-05	Data migration appears late (Q-15)	Medium	Decide at P0; import tools if required
RK-06	Client isolation defect	High	Server-side enforcement and automated tests
RK-07	Ledger and balance drift	High	Append-only design and nightly reconciliation
RK-08	Volumes exceed assumptions (Q-14)	Medium	Confirm figures, load test at P1

20.2 Open questions

All 24 questions in the Part 1 analysis (Q-01 to Q-24) remain open. Priority is unchanged: Q-01 to Q-06 block architecture and data model and should be answered before design freeze. Every requirement marked PCA changes status only after the client's written answer.

21. Handover

Deliverable	Content
Source code and repository	Application, migrations, adapters, tests
Environment guide	Setup, configuration, secrets handling, deployment steps
Database documentation	ERD, data dictionary, ledger rules
Integration guide	eBay setup, token renewal, sync operation, exception handling
Admin and user guides	By role, with workflow screenshots
Test evidence	Test plan, results, acceptance sign-off per phase
Operations runbook	Backups, restore test, monitoring, incident steps
Seed data	Clearly labeled demo data and the script to remove it

Final quality note

This SRS covers all 21 sections of the target structure. Its weakest areas are the ones the client left open: ownership model, allocation rule, valuation, VAT and currency, storage billing, volumes and retention. Each is labeled PCA, and the document should be re-issued as v1.0 after the P0 questions are answered.

Appendix A. Part 2 Detail: Document Control, Business Context, Scope and Roles

This appendix is the detailed Part 2 from the roadmap. It deepens Sections 1 to 4 and does not replace them. Where the two differ, this appendix governs once approved. All items are PCA unless stated.

A.1 Document control detail

A.1.1 References

Ref	Document	Use
REF-01	Client requirement document (30 sections)	Source of all requirements
REF-02	Part 1: Requirement Analysis, Coverage Checklist, Client Questions	Gaps, assumptions, Q-01 to Q-24
REF-03	Current official eBay developer documentation	To be verified before integration design (D-01)
REF-04	Applicable VAT and data-protection rules	To be confirmed with the client (Q-08, Q-20)

A.1.2 Stakeholders and approvals

Role	Responsibility	Sign-off needed for
Business owner (client)	Answers P0 questions, approves scope and phases	SRS v1.0, phase gates
Operations lead (client)	Confirms workflows, status model, allocation rule	S6, S7
Finance lead (client)	Confirms valuation, VAT, currency, cost model	S12
Warehouse lead (client)	Confirms locations, storage basis, returns handling	FR-WH, FR-STO, FR-RET
Project analyst	Maintains this SRS and the traceability matrix	Change log
Technical lead (developer)	Confirms feasibility, adapters, NFR targets	S9 to S11, S13
QA lead (developer)	Maps acceptance criteria to test cases	S17, S18

A.1.3 Document status and change control

- Status path: Draft, In review, Approved. Version 1.0 is issued only after Q-01 to Q-06 are answered in writing.
- A requirement tagged PCA moves to Approved when the client's written answer is attached to the matching question.
- After approval, changes need a change request with impact on scope, data model and phase, and a new version number.
- The traceability matrix (S18) is updated with every change.

A.2 Business context detail

A.2.1 Ownership scenarios (decision pending, Q-01)

Scenario	Stock owner	eBay accounts	Effect on design
A. Own stock only	Operator	Operator accounts	Client isolation unused but still built into the schema
B. 3PL only	Clients	Client accounts or manual orders	Per-client eBay connections and strict isolation from P1
C. Both (assumed)	Operator and clients	Mixed	Owner on every stock, order and charge record; client eBay connections optional

The schema in S10 supports all three. The choice changes only configuration, the P1 UI and the order of work in P3.

A.2.2 Business processes in scope

Process	Start	End	Modules
Order to cash (eBay)	Order imported	Order delivered, revenue recognized	FR-EBAY, FR-ORD, FR-INV, FR-SHIP
Procure to pay	PO drafted	Supplier paid	FR-SUP, FR-PO, FR-PAY
3PL service to cash	Inbound notice	Client invoice paid	FR-3PL, FR-FUL, FR-STO, FR-INVC

Process	Start	End	Modules
Returns	Return opened	Disposition and refund recorded	FR-RET
Stock control	Receipt or adjustment	Reconciled balance	FR-INV, FR-ADJ, FR-WH
Record to report	Any posting	Reconciled report	FR-RPT, FR-PRF, S12

A.2.3 Success criteria for the first release (P1)

- All eBay orders from connected accounts are visible and traceable to their raw data.
- Stock balances can be explained from the ledger for any SKU and date.
- No order can reserve or ship stock that does not exist.
- Every change to stock and orders is in the audit log.

A.3 Scope detail per phase

In means built and tested in the phase. Hidden means the screen or widget does not appear until the phase.

Module	P1	P2	P3	P4	P5
Auth, roles, audit	In				
eBay import	In				Write-back (if approved)
Orders	In		Client orders		
Products, SKUs	In	Bundles (if confirmed)			
Inventory, ledger	In (one default location)	Bin locations, valuation	Client stock views		
Adjustments	In	Cycle counts			
Suppliers, POs	Hidden	In		Supplier payments	
Returns	Hidden	In		Refund postings	
Customers	Hidden	In			

Module	P1	P2	P3	P4	P5
3PL, fulfilment, storage	Hidden	Hidden	In		
Invoices, payments, expenses, profit	Hidden	Hidden	Hidden	In	
Dashboard	Operational KPIs	Inventory value	Client dashboard	Finance KPIs	
Reports	R-01 to R-06	R-07 to R-09	R-10, R-11	R-12 to R-14	Advanced
Notifications	In-app indicators				Engine, email
Shipping	Manual entry				Carrier APIs

A.3.1 Phase dependencies

Phase	Depends on
P1	D-01 (eBay access), D-02 (allocation rule), D-03 (SKU mapping), D-04 (ownership model)
P2	D-05 (valuation) for inventory value; Q-05 for warehouse scope
P3	D-04, D-06 (storage billing rules)
P4	D-05 (valuation, VAT, currency), Q-09, Q-12
P5	D-07 (carrier terms), Q-07 (write-back decision), Q-22 (channels)

A.3.2 Scope boundaries

- The platform records payments but does not process them (Q-18).
- It is not an accounting package. It produces a balanced ledger and exports for an external accountant; automated accounting sync is future (client Section 29).
- Multi-company is out of scope. Multi-warehouse and multi-eBay-account are stored in the schema but shown in the UI only if Q-03 and Q-05 confirm them.

A.4 Role detail

A.4.1 Role profiles

Role	Typical user	Data visibility	Cannot do
Super Admin	Owner, IT lead	All data, all owners	Approve own adjustments or expenses
Operations Manager	Head of operations	All operational data, costs visible	Change system configuration or users
Warehouse Staff	Pickers, packers, receivers	Orders, stock and locations; no costs, margins or finance	Approve adjustments, see invoices or profit
Finance User	Accountant	Finance data and read access to operations	Change stock, orders or locations
3PL Client User	Client employee	Own client data only	See other clients, operator costs, or internal margins
Viewer	Auditor, director	Read access to permitted modules	Create, update, approve or export restricted data

A.4.2 Approval authority matrix (PCA, Q-13)

Action	Requester	Approver	Threshold
Stock adjustment	Warehouse, Ops Manager	Ops Manager or Super Admin, never the requester	Configurable quantity or value
Purchase order	Ops Manager	Super Admin	Configurable value
Expense	Finance User	Super Admin	Configurable value
Invoice void or reversal	Finance User	Super Admin	Always, with reason
Refund	Ops Manager	Finance User	Configurable value
Configuration change	Super Admin	Logged, second Super Admin if two exist	Always audited

A.4.3 Separation of duties

- The person who creates a stock adjustment, expense or PO cannot approve it.
- Finance cannot post stock movements and Warehouse cannot post financial documents.
- Super Admin actions are the most heavily audited, and exports of financial or personal data are always logged.

A.4.4 Client user rules (P3)

- Each client user belongs to exactly one client, and the server adds the client filter to every query. The UI filter is never the only protection.
- Client users cannot see the operator's own stock, other clients' SKU names, or internal costs.
- Client invoices and storage statements are visible only after they are issued.
- Client accounts are created by the operator and can be disabled without deleting history.